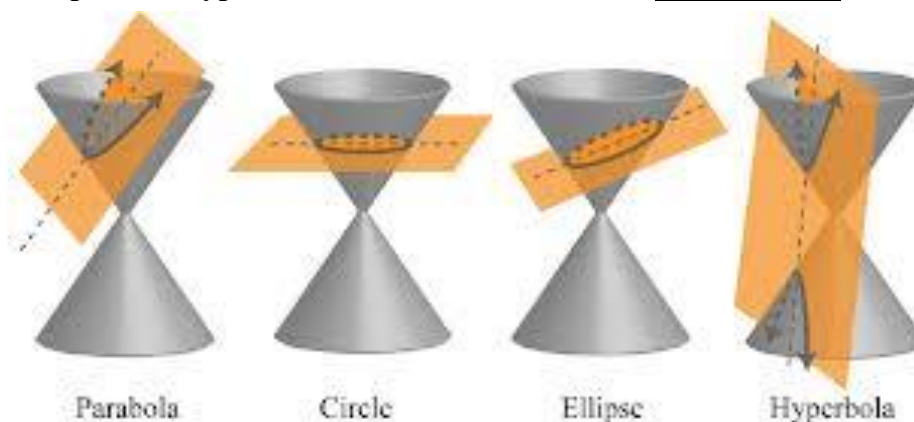


5.4 Summary of Conic Sections

If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.



The Equation of a Parabola: (opening up or down)

The parabola

$$y = a(x - h)^2 + k$$

has focus $(h, k + p)$

and directrix $y = k - p$

where $a = \frac{1}{4p}$ and (h, k) is the vertex.

The Equation of a Parabola: (opening left or right)

The parabola

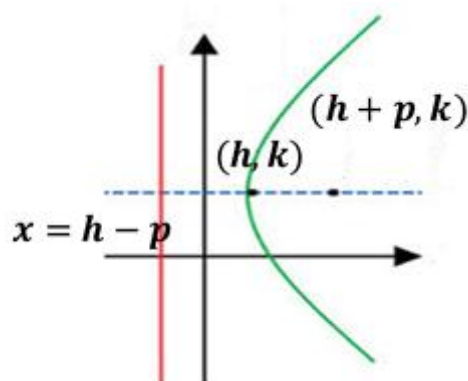
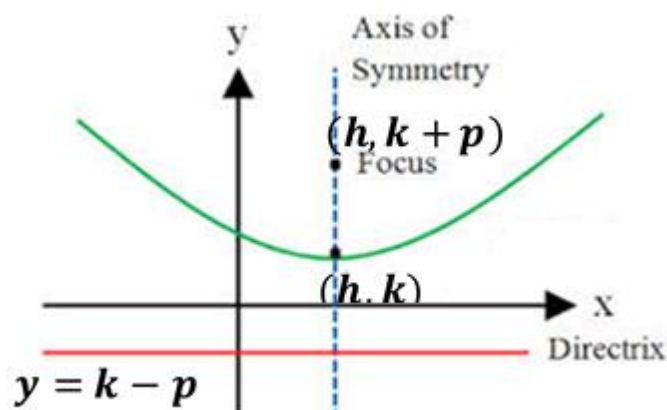
$$x = a(y - k)^2 + h$$

has focus $(h + p, k)$

and directrix $x = h - p$

where $a = \frac{1}{4p}$ and (h, k) is the vertex.

Note: If $a > 0$ then the parabola opens to the right. If $a < 0$ then the parabola opens to the left..



The Equation of an Ellipse: (with a *horizontal* major axis and center at the origin)

The ellipse with center at the origin and equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b$$

has vertices $(\pm a, 0)$, endpoints of the minor axis $(0, \pm b)$, and foci $(\pm c, 0)$.

where $c^2 = a^2 - b^2$

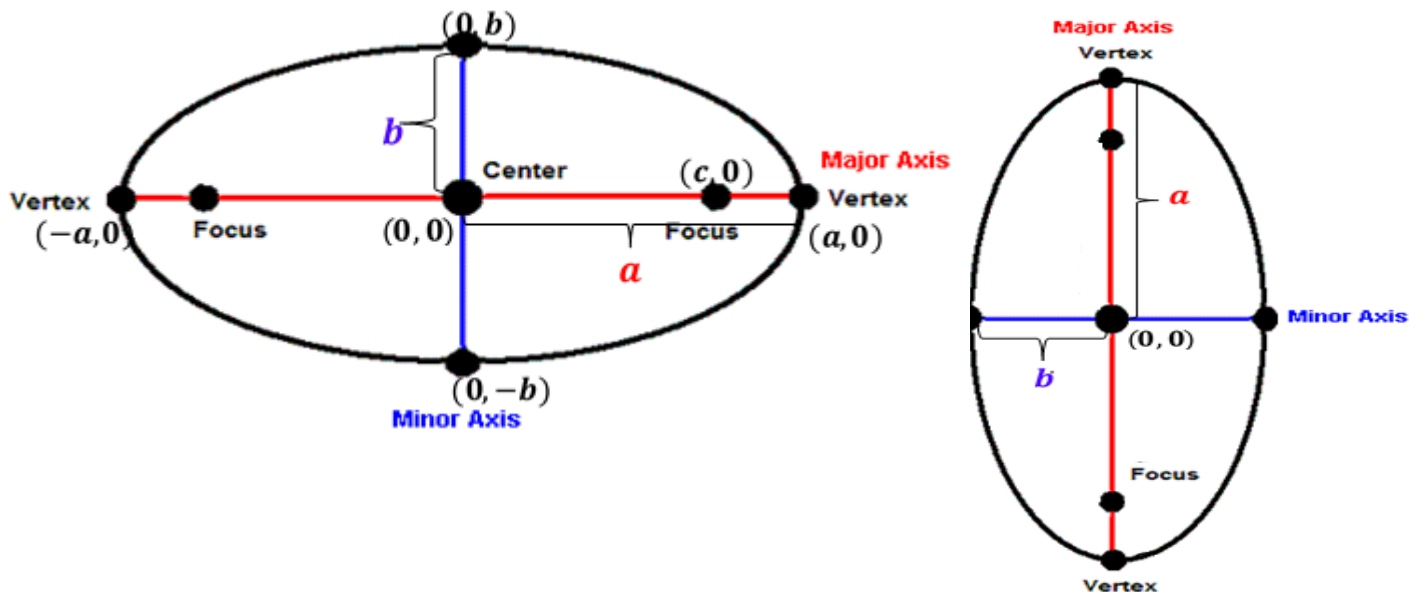
The Equation of an Ellipse: (with a *vertical* major axis and center at the origin)

The ellipse with center at the origin and equation

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1, \quad a > b$$

has vertices $(0, \pm a)$, endpoints of the minor axis $(\pm b, 0)$, and foci $(0, \pm c)$.

where $c^2 = a^2 - b^2$



Equation for a Circle

The standard equation of a circle with center (h, k) and radius r ($r > 0$) is

$$(x - h)^2 + (y - k)^2 = r^2$$

Note: By dividing both sides of this equation by r^2 we obtain $\frac{(x-h)^2}{r^2} + \frac{(y-k)^2}{r^2} = 1$. So clearly a circle is an ellipse with major and minor axes of the same length.

The Equation of a Hyperbola: (opening left or right)

The equation of a hyperbola with center at the origin with foci $(\pm c, 0)$ and x -intercepts $(\pm a, 0)$ is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where $c^2 = a^2 + b^2$

The differences of the distances from any point of the hyperbola to the two foci is $2a$.

The Equation of a Hyperbola: (opening up or down)

The equation of a hyperbola with center at the origin with foci $(0, \pm c)$ and y -intercepts $(0, \pm a)$ is

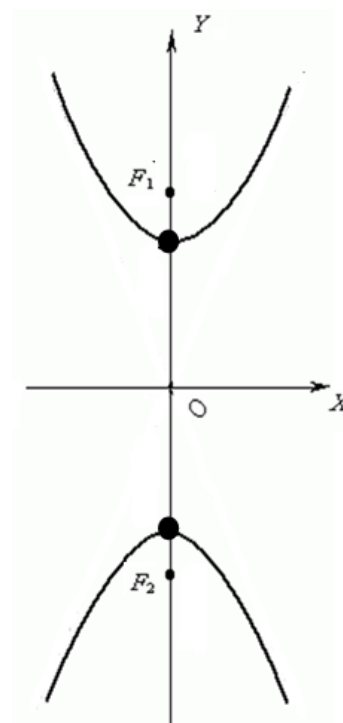
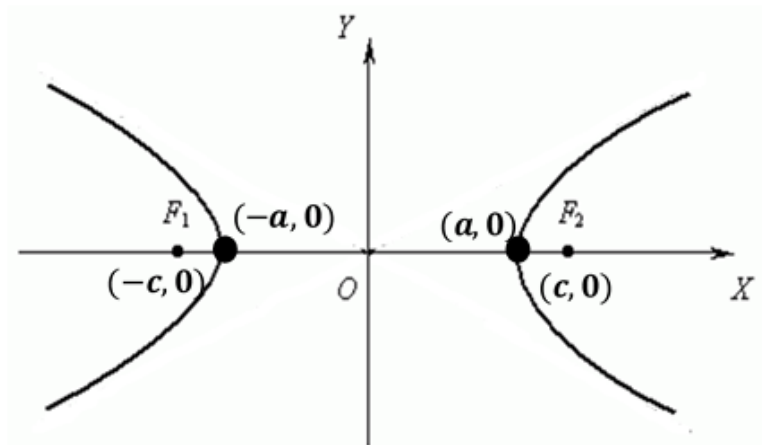
$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

where $c^2 = a^2 + b^2$

The differences of the distances from any point of the hyperbola to the two foci is $2a$.

Note: With hyperbolas it is not important whether a or b is the larger value. We always think of a as the value beneath y and b as the value beneath x .

Note: If $a > 0$ then the parabola opens up. If $a < 0$ then the parabola opens down.

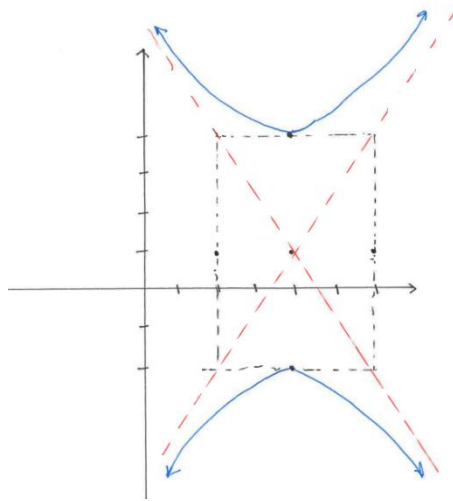


Example 1: Determine what type of conic section represented by the equation $9x^2 - 4y^2 - 72x + 8y + 176 = 0$ and sketch a rough graph of the conic.

We must complete the square in order to get the equation in a form that we can recognize.

$$\begin{aligned} 9x^2 - 4y^2 - 72x + 8y + 176 &= 0 \Leftrightarrow 9x^2 - 72x - 4y^2 + 8y = -176 \\ \Leftrightarrow 9(x^2 - 8x) - 4(y^2 - 2y) &= -176 \Leftrightarrow 9(x^2 - 8x + 16) - 4(y^2 - 2y + 1) = -176 + 144 - 4 \\ \Leftrightarrow 9(x-4)^2 - 4(y-1)^2 &= -36 \Leftrightarrow \frac{(y-1)^2}{9} - \frac{(x-4)^2}{4} = 1. \end{aligned}$$

From this we recognize that the equation determines a hyperbola which opens vertically and has center $(4, 1)$.

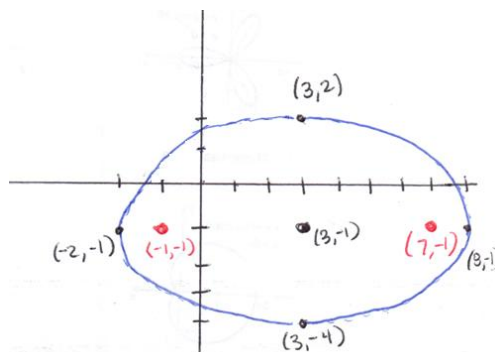


Example 2: Determine what type of conic section represented by the equation $9x^2 - 54x - 19 = 100 - 50y - 25y^2$ and sketch a rough graph of the conic.

$$\begin{aligned} 9x^2 - 54x - 19 &= 100 - 50y - 25y^2 \Leftrightarrow 9x^2 - 54x + 25y^2 + 50y = 119 \Leftrightarrow 9(x^2 - 6x + \quad) + 25(y^2 + 2y + \quad) = 119 \\ \Leftrightarrow 9(x^2 - 6x + 9) + 25(y^2 + 2y + 1) &= 119 + 81 + 25 \\ \Leftrightarrow 9(x-3)^2 + 25(y+1)^2 &= 225 \\ \Leftrightarrow \frac{(x-3)^2}{25} + \frac{(y+1)^2}{9} &= 1 \end{aligned}$$

$$a^2 = 25, \quad b^2 = 9 \quad c^2 = 25 - 9 = 16, \quad c = \pm 4$$

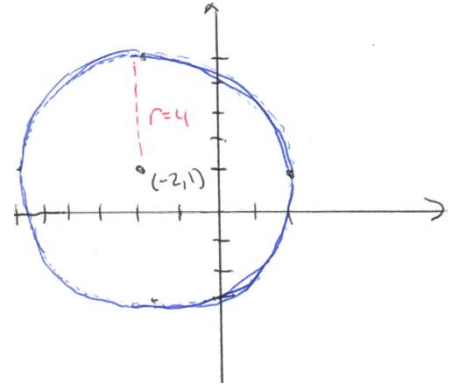
The center of the ellipse occurs at the point: $(3, -1)$



Example 3: Determine what type of conic section represented by the equation $x^2 + 4x + y^2 - 2y = 11$ and sketch a rough graph of the conic.

$$\begin{aligned} (x^2 + 4x + \underline{\quad}) + (y^2 - 2y + \underline{\quad}) &= 11 \\ \Rightarrow (x^2 + 4x + 4) + (y^2 - 2y + 1) &= 11 + 4 + 1 \\ \Leftrightarrow (x + 2)^2 + (y - 1)^2 &= 16 \end{aligned}$$

So we have a circle with center $(-2, 1)$ and radius $r = \sqrt{16} = 4$.



Example 4: Determine what type of conic section represented by the equation $2y + x - y^2 = 0$ and sketch a rough graph of the conic.

$$x = y^2 - 2y \quad \Leftrightarrow \quad x = y^2 - 2y + 1 - 1 \quad \Leftrightarrow \quad x = (y - 1)^2 - 1$$

Clearly this is a quadratic that opens right and has vertex $(-1, 1)$.

